Persuasion via Man and Machine

A Living Review

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A Living Review

Living Review · Draft

This is a **living review** — entries are continuously added and revised as the field evolves. The LaTeX source, bibliography, and compiled PDF are always available on GitHub. Readers are encouraged to open an issue to suggest missing work.

Also see the author's PhD thesis — *Behavior As A Modality: A Framework To Enable Automated Persuasion* — for closely related work.

1.1 Abstract

This living review synthesises the science and engineering of persuasion across five millennia of scholarship and across the disciplinary boundaries that have kept the field fragmented — rhetoric, psychology, linguistics, political science, marketing, and artificial intelligence. We argue that persuasion is best understood as the optimisation of the effect of a message on a receiver, and that machine learning is now a fully-fledged participant in that optimisation — as an instrument, as an object of study, and as a threat.

Because this is a living review, entries are added and revised continuously. Readers are encouraged to open an issue on GitHub to suggest missing work.

1.2 How to Cite

```
@misc{singla2025persuasion,
  author    = {Singla, Yaman Kumar},
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```

2 Introduction

Before there was language, there was persuasion. Long before *Homo sapiens* assembled in agoras to debate, before Aristotle codified rhetoric, before the first written argument was pressed into clay, living organisms were influencing one another's behaviour through chemical gradients, postural displays, and acoustic signals. Persuasion — understood as the capacity to alter the beliefs, states, or actions of another agent — is not a cultural invention. It is a biological imperative, as old as sociality itself.

This matters for understanding what persuasion *is*, because it reveals the function persuasion serves at its deepest level. Across the tree of life, wherever organisms face the challenge of coordinating their actions — foraging, reproducing, defending territory, allocating labour — natural selection has repeatedly converged on communication systems that allow one individual to change what another does. In eusocial insects, the waggle dance of a returning forager persuades thousands of nestmates to fly to a precise location they have never visited. In primates, grooming and vocal signalling maintain cooperative alliances within groups. In humans, language extends this logic to a scale and abstraction that no other species has achieved: shared narratives, institutions, and markets are all, at their foundation, persuasive achievements maintained by the ongoing communicative activity of their participants.

Persuasion, in this view, is the mechanism by which cooperation at scale becomes possible — and cooperation at scale is what has allowed *Homo sapiens* to dominate every habitat on Earth. We return to this evolutionary perspective in depth in Section 3.1. For now, it is enough to note that the study of persuasion is not the study of a human peculiarity. It is the study of one of evolution's most powerful and recurrent inventions.

Communication is defined as optimising the effect of a message on a receiver sent over a channel by a sender at a particular time [20, 21, 36]. This optimisation of altering the receiver's opinions, beliefs, or actions is also referred to as **persuasion**. The study of communication or persuasion as a universal human practice is among the most ancient of human concerns. Persuasion has been looked at as the art and science of how we get anything done. Once, it helped our ancestors to plan and execute hunting expeditions, and today, it helps us plan massive projects like the Large Hadron Collider and even inter-planetary expeditions like Europa Clipper.

Persuasion enables making ourselves understood sufficiently well so that we can coordinate our actions. The ability to persuade is not unique to our species. Persuasion is observed in both conspecific [13] and interspecific [16] scenarios. For example, both monkeys and apes yielded episodes indicating they were able to judge very finely how to obtain or hide desirable objects deceptively from the gaze of others [13]. We humans are exceptional in our capacity to cooperate with strangers. Language allowed our ancestors to cooperate, and helped to resolve conflicts by exchanging information, though this includes invented fictions, social constructions, and other imagined realities.

For example, on encountering the explorer James Cook, two Fuegian islanders stepped forward to display and then throw aside large sticks, gestures that Cook interpreted as indicating peaceful intentions. Indeed, the Fuegians and Cook were soon exchanging gifts and eating together on board ship.

4 years were children very competent in recognizing that others' mental representations of the world could be very different to their own and need to be computed to predict their future actions accurately, as seen most revealingly in the context of deception and false beliefs (Wimmer & Perner 1983).

A second, and no less important, reason involves influencing other Homo sapiens (Mercier & Sperber, 2011). Sloman and Fernbach (2017) assure us of the very real limits of our individual knowledge and its inadequacy for accomplishing many, perhaps most, of the tasks that we face daily.

One of the oldest books on persuasion, called *Precepts*, written by Ptah-Hotep for the Pharaoh, is 4,500 years old (2375 BCE). Rhetoric — the use of symbols to persuade other humans — began in ancient Greece and its Mediterranean colonies in the fifth century BCE when there were situations of collective decision making that inspired some practitioners to ask how persuading others was best accomplished.

The study of persuasion as a discipline is in its third millennium. It was Plato's student, Aristotle, who provided the first comprehensive theory of rhetorical discourse. He defined rhetoric in terms of "observing in a given case the available means of Persuasion," instructing that the "available means" encompassed a range of appeals, some grounded in logic (*logos*), others in emotion (*pathos*), and still others in the communicator (*ethos*). Aristotle urged communicators to base judgments about the most appropriate means of Persuasion on the nature of the audience. His views had long-lasting impact: "Aristotle's theory of rhetorical discourse has withstood the test of time, furnishing axioms that guide today's practitioners of Persuasion and campaigns."

In the West, the tumultuous emergence of democracy in fourth-century BCE Athens gave rise to a class of civic intellectuals teaching persuasive speaking — known as Sophists — to the establishment of rhetoric as a field of study, and to clashes with another new field, philosophy, over explanations of the linguistic features running through belief, knowledge, argumentation, and human conduct. The early modern period was another epoch of intensified interest, with the rise of print, the Reformation, technoscience, and the colonial pursuits of Europe. The middle of the twentieth century was still another, when propaganda studies, argumentation studies, and the New Rhetoric all arose from the trauma of the Second World War.

We are in another such period now. Large Language Models have opened a new chapter in the history of persuasion — one in which the sender of a message may be a machine, the message may be tailored to a single individual at a cost approaching zero, and the channel may be every screen simultaneously. This review documents what we know, what we do not, and what the field must now answer.

> "Persuasion is ubiquitous in the political process; it is also the central aim of political interaction. It is literally the stuff of politics."
>
> — Mutz, Sniderman & Brody, 1996

When people think of persuasion they often think of debates over controversial topics, where distinct individuals and groups invested in different viewpoints confront one another. In these situations, successful persuasion amounts to convincing people to cross over from one side to the other. However, most fully formed persuasive discourse is not aimed at people who disagree anyway; in fact it is usually aimed at people who already agree, but whose agreement can be lessened or intensified.

Persuasion is also aimed at forming beliefs in the first place in people who have no stance on an issue at all. Response shaping is roughly the acquisition of an attitude, whereas response reinforcement can be equated with strengthening a preexisting attitude. By contrast, response changing references movement across the midpoint of an attitude scale.

At both a more subtle and a more powerful level, it has been argued that the Persuasion profession "serves not so much to advertise products as to promote consumption as a way of life" (Lasch, 1978, p. 72). But many of these same marketing techniques are also used to help solve pressing social problems such as improvement of the nation's health.

Case Study: "Don't Mess With Texas"

In 1985, "Don't Mess With Texas" bumper stickers began appearing on cars in Texas, beginning the launch of what turned out to be the most successful anti-littering campaign ever conducted in the United States. The campaign, commissioned by the Texas Department of Transportation, targeted young male truck drivers. The advertising team recruited well-known masculine icons — members of the Dallas Cowboys, Willie Nelson, Matthew McConaughey — who looked sternly into the camera as they crushed beer cans and proclaimed the slogan.

The campaign capitalised on Texan pride, reduced litter by 72% in six years, and is now a textbook example of **message–audience fit** in persuasion.

Charismatic leaders such as Nelson Mandela, Martin Luther King Jr., and Mahatma Gandhi have stirred the masses and brought about radical social change, even when lacking significant institutional power or money. And in 2014, the "Ice Bucket Challenge" persuaded people to donate, ultimately taking in over $115 million, funding groundbreaking ALS research.

People, however, can be remarkably resistant to persuasion. Many well-designed, well-funded efforts to encourage people to practice safe sex, stop using drugs, or improve their diet have failed [30]. People can be stubbornly resistant to changing their minds, even when their health or economic well-being is affected.

2.1 Why This Survey?

The basic difference between the sciences that study persuasion can be characterised as: the *how* of persuasion vs the *what* of persuasion. Right now, machine learning is not given a seat at the table where rhetoric and persuasion are discussed. Top academic programmes in rhetoric do not discuss the advances that computers have enabled.

Similarly, in the machine learning field, there has been comparatively little research on persuasion. Further, whatever research there has been has been in the context of optimising content for persuasion. On the other hand, persuasion should be seen as optimising across the *source*, *channel*, *time*, *message*, and even the *audience* — the full **SCEMA** framework.

What makes the study of persuasion interesting and difficult at the same time is that every field has tried to answer the same fundamental question from a different vantage point, accumulating incompatible vocabularies, incommensurable measures, and siloed literatures. This review attempts a unification.

2.2 Early Beginnings

- Language itself is believed to have developed largely because of the need for cooperation — therefore, persuasion is coextensive with sociality.
- Aristotle's *Rhetoric* (4th c. BCE) remains the founding document of the field.
- Mythology is an artefact of persuasion: narratives that mobilise groups around shared values.
- Linguistic devices — tropes, metaphors, humour — are the micro-mechanisms through which persuasion operates.

2.3 Common Misconceptions and Ethical Challenges

> Caution
>
> This section is under active development. Suggest content via GitHub Issues.

3 Persuasion as a Major Transition in Evolution

> **Chapter Overview**
>
> This chapter situates persuasion within deep evolutionary history. We argue that the capacity to influence the beliefs and actions of others is not merely a cognitive or cultural phenomenon — it is a force that has repeatedly restructured life at the level of the organism, the colony, and the species.

3.1 The Major Transitions Framework

In their landmark work, [25] identified a small number of pivotal moments in the history of life when the unit of selection shifted upwards — when formerly independent replicators became integrated into a higher level of organisation. These *major transitions* include the origin of the eukaryotic cell from independent prokaryotes, the emergence of multicellular organisms from single cells, and the origin of eusocial insect colonies from solitary ancestors. What each transition shares is a common underlying logic: individual interests become subordinated to a collective, and the mechanisms that enforce this subordination are fundamentally communicative — signals, cues, and the suppression of defection.

We propose that **persuasion — the capacity to alter the beliefs, states, or actions of another agent — is the generative mechanism behind these transitions.** Coordination at scale requires not just the ability to act, but the ability to make others act. In this light, the history of persuasion is inseparable from the history of biological complexity.

3.2 Eusociality: Persuasion at the Colony Level

The most dramatic example of persuasion enabling a major evolutionary transition is eusociality — the organisation of insect societies such as honeybees (*Apis mellifera*), leafcutter ants (*Atta*), and termites (*Macrotermes*) into colonies in which most individuals forgo personal reproduction entirely in order to serve the collective. This is, by any measure, one of the most radical subordinations of individual interest in the history of life.

3.2.1 How Bee Colonies Coordinate

Honeybee colonies achieve coordinated collective action through a rich communication system. Seeley [35] documented at least seventeen distinct signal types exchanged between colony members — chemical and mechanical — covering brood recognition, queen presence, aggression, and forager recruitment. The celebrated **waggle dance**, first decoded by Karl von Frisch [9], encodes the direction and distance of a food source relative to the sun with extraordinary precision: the angle of the dance from vertical corresponds to the azimuthal angle of the food source from the sun, and the duration of the waggle run encodes distance. A forager returning to the hive does not merely signal "food exists" — it delivers a spatial argument that persuades nestmates to fly to a specific location they have never visited.

This is persuasion in a precise sense: a sender encodes information about the world that changes the behavioural dispositions of receivers, who act on that information. The mechanism is sophisticated enough that colonies can democratically select new nest sites through a process of competitive recruitment dances, in which scouts advocate for competing sites until a quorum is reached [34]. The colony's decision emerges from a form of deliberation — a process structurally analogous to human debate.

3.2.2 Division of Labour and the Suppression of Defection

The stability of the colony depends on the suppression of individual reproductive ambition. In honeybees, workers are capable of laying unfertilised (male) eggs, but they rarely do so because **worker policing** is enforced by nestmates who identify and destroy worker-laid eggs using chemical recognition cues [32]. This enforcement is itself a communicative act — a signal about what is and is not sanctioned within the collective.

Task allocation is regulated physiologically by juvenile hormone (JH), which rises with age, shifting workers from brood care to foraging [33]. As the colony's needs change — a dearth of foragers, or a surplus of brood — the pace of this hormonal transition adjusts dynamically, mediated by chemical signals from the brood and from the queen. The queen's pheromones simultaneously suppress worker ovarian development and maintain colony cohesion: they are not merely chemical signals but acts of continuous persuasion, sustaining the colony's cooperative structure moment to moment.

3.2.3 Ants: Chemical Language at Scale

Ants (*Formicidae*) have independently evolved eusociality many times and have colonised virtually every terrestrial habitat on Earth. Their success rests on a chemical communication system of remarkable elaboration. Leafcutter ants (*Atta cephalotes*) lay pheromone trails from nest to food source that encode not only direction but, through the concentration and composition of the trail, information about food quality and quantity [40]. When a food source is depleted, ants returning empty-handed deposit less pheromone, allowing the trail to dissipate — an automatic down-regulation of recruitment that prevents the colony from committing further effort to an exhausted resource. The information propagated through the pheromone network is, in effect, a distributed argument about where collective effort should be directed.

Army ants (*Eciton burchellii*) build living bridges from their own bodies to span gaps in the forest floor, with the decision of when to disassemble the bridge emerging from local traffic information — each ant in the bridge responds to whether nestmates are walking over it. No individual plans the bridge; the structure is the emergent result of thousands of local persuasive acts.

3.2.4 Kin Selection and the Logic of Cooperative Persuasion

Why would an individual evolve to be persuadable in ways that benefit others at a cost to itself? [11] showed that altruistic behaviour could evolve when it benefits sufficiently close relatives, formalised as Hamilton's rule: $rb > c$, where r is genetic relatedness, b is the benefit to the recipient, and c is the cost to the actor. In Hymenoptera (bees, ants, wasps), haplodiploidy — the mechanism by which females develop from fertilised eggs and males from unfertilised ones — means that full sisters share three-quarters of their genes, a relatedness higher than that between a mother and her offspring [38]. This elevated relatedness increases the inclusive fitness benefit of helping the colony, making the evolutionary ground fertile for a communication system in which receivers reliably act on senders' signals.

This connection between kinship and cooperability is not merely academic: it suggests that the *trustworthiness* of a persuasive signal — the probability that a receiver will act on it — is itself subject to evolutionary selection. Signalling systems evolve toward honesty when sender and receiver interests are sufficiently aligned, and toward manipulation when they diverge. The evolutionary tension between honest communication and deceptive signalling is the deep structure underlying all persuasion [16].

3.3 Language: The Human Major Transition

The emergence of human language represents an analogous, though culturally rather than genetically transmitted, major transition in the organisation of cooperation. Language is unique among animal communication systems in its **combinatorial productivity** — the capacity to express an unbounded number of distinct meanings from a finite set of elements — and in its ability to communicate about **displaced referents**: objects, events, and agents that are not present in the immediate environment [14].

This displacement property is what makes language an instrument of social organisation at scales unattainable by any other species. A honeybee's waggle dance can coordinate foraging across a few kilometres. Human language can coordinate building the Large Hadron Collider across a hundred nations. The mechanism is the same — a sender encodes information that changes the behavioural dispositions of receivers — but the representational power of language makes it possible to communicate about abstract entities: rules, roles, obligations, futures, and fictions.

3.3.1 Shared Fictions as Coordination Technology

[12] argued that what distinguishes *Homo sapiens* from other animals is not superior individual intelligence but the ability to cooperate flexibly in large groups through **shared myths** — collectively believed fictions that coordinate behaviour. Money, nations, corporations, and religions are all persuasive constructs: they exist because enough people believe they exist, and that belief generates coordinated behaviour. In this sense, every institution is a persuasive achievement — maintained not by physical force but by the ongoing, distributed persuasion of its members that the institution is real and worth sustaining.

This is the deepest sense in which persuasion enables societies: not merely by changing individual minds but by constructing and maintaining the shared representational structures within which social action becomes possible.

3.3.2 Language, Gossip, and the Maintenance of Cooperation

[8] proposed that language evolved primarily as a form of **social grooming** — a way of maintaining cooperative alliances among individuals who could not physically groom all their allies simultaneously. In small primate groups, physical grooming is the primary mechanism for building and maintaining social bonds. As group sizes grew beyond the capacity for physical grooming (estimated at roughly 50 individuals), language allowed the same bonding function to be performed at a distance and in parallel — one individual could simultaneously maintain relationships with many others through conversation.

A critical component of this social maintenance function is **reputation management through gossip**: exchanging information about the cooperative or defecting behaviour of third parties [26]. Reputational persuasion — communicating to others that a particular individual is trustworthy or untrustworthy — is a powerful mechanism for enforcing cooperation in large groups without centralised authority. The group's collective persuasive acts (gossip, praise, censure) police free-riding in a manner structurally parallel to worker policing in honeybee colonies.

3.4 Persuasion Across Scales

The argument developed here suggests that persuasion is not merely a feature of human culture but a **fundamental biological mechanism for enabling cooperation at successively larger scales.** Each major transition in the organisation of cooperation has been enabled by a new communication technology:

Scale	Organism	Communication Technology	Transition Enabled
Cell → organism	Multicellular animals	Chemical signalling (hormones, receptors)	Differentiation and division of labour
Individual → colony	Eusocial insects	Pheromones, mechanical signals (dance)	Superorganism
Band → tribe	Early *Homo*	Proto-language, gesture	Coordinated hunting, territory defence

Scale	Organism	Communication Technology	Transition Enabled
Tribe → civilisation	*Homo sapiens*	Symbolic language, writing	Institutions, states, markets
Nation → globe	Modern humans	Print, broadcast media, internet	Global coordination
Human → AI	Emerging	Large language models	*To be determined*

The final row in this table is the subject of this review. LLMs represent a new kind of communicative agent — one capable of generating persuasive messages at industrial scale, tailored to individual recipients, across every channel simultaneously. Whether this constitutes a new major transition in the organisation of cooperation, or a profound disruption of existing cooperative structures, is the central unanswered question of our moment.

Language and Persuasion

The connection between language and persuasion is taken up in detail in **?@sec-views**, where we examine how different disciplines — linguistics, social psychology, economics, neuroscience — have conceptualised the mechanisms by which messages change minds. The evolutionary framing developed here provides the deepest context for why those mechanisms exist at all.

4 Views of Different Fields

Persuasion is one of those rare phenomena that every human discipline claims as its own. The table below maps the dominant lens, the canonical measure, and the key open question that each field brings to the common problem.

Table 4.1: Overview of disciplinary perspectives on persuasion

Field	Dominant Lens	Key Construct
Psychology	Attitude change, dual-process	ELM, dissonance
Linguistics	Phonology, syntax, framing	Powerless speech, hedging
Sociology	Social norms, networks	Diffusion, conformity
Economics	Incentives, biases	Nudge, loss aversion
Political Science	Propaganda, agenda	Framing effects
Marketing	Ad effectiveness	Memorability, recall
CS & AI	NLP, argumentation, LLMs	Claim detection, stance
Neuroscience	Neural correlates	fMRI of attitude change

4.1 Psychology

Psychology has produced the most influential theoretical frameworks in persuasion research. Three dominate the modern literature:

Cognitive Dissonance Theory [festinger1957theory?] People experience discomfort when holding inconsistent beliefs and are motivated to reduce it, making them susceptible to persuasion that resolves the inconsistency.

Matching Hypothesis Messages are more persuasive when they match the audience's regulatory focus (promotion vs. prevention orientation).

Elaboration Likelihood Model [30] The dual-process model distinguishing the *central route* (argument-quality driven) from the *peripheral route* (heuristic driven). The route taken depends on motivation and ability to process the message.

4.2 Linguistics

Hosman neatly subdivides the linguistics of persuasion in terms of *phonology*, *syntax*, *lexicon*, and *text/narrative* before turning to an examination of the effects of each on judgements of the message source, recall, and attitude change. Key findings include: powerful versus powerless speech styles, hedging, intensifiers, rhetorical questions, and the surprising persuasive force of narrative over argument in some populations.

4.3 Sociology

> Caution
>
> Section under active development.

4.4 Economics & Behavioural Economics

Behavioural economics treats persuasion through the lens of systematic cognitive biases — the anchoring effect, loss aversion, default effects, and social proof — that make humans predictably susceptible to certain message architectures. Thaler & Sunstein's *nudge* framework formalised this: changing the choice architecture can shift behaviour without mandate, a form of structural persuasion.

4.5 Political Science

> "Persuasion is ubiquitous in the political process; it is also the central aim of political interaction. It is literally the stuff of politics."
>
> — Mutz, Sniderman & Brody, 1996

Political science studies persuasion at the level of populations: propaganda, agenda-setting, framing effects, and the conditions under which people update political beliefs. Electoral persuasion effects are famously small but real — and AI micro-targeting threatens to amplify them at industrial scale.

4.6 Marketing & Advertising

Marketing operationalises persuasion at commercial scale. The discipline has produced rich empirical literatures on message memorability, the effectiveness of emotional vs. rational appeals, celebrity endorsement, scarcity and social proof, and the varying persuasive power of different media channels. The rise of programmatic advertising and data-driven micro-targeting has brought marketing into direct dialogue with AI.

4.7 Computer Science & AI

CS and AI engage persuasion through multiple sub-fields:

- **Formal argumentation** — Dung's abstract argumentation frameworks, computational models of argument strength.
- **Argumentation mining in NLP** — automatic identification of claims, premises, and attack/support relations in text.
- **Computer vision** — recognising persuasive intent in images and video; visual rhetoric; ad-effectiveness prediction.
- **Large Language Models** — generating and detecting persuasive text; the frontier where the field is now accelerating most rapidly.
- **Agentic systems** — AI agents that negotiate, persuade, and adapt their communication strategy to an interlocutor.

4.8 Neuroscience

> Caution
>
> Section under active development.

5 Types of Work in Automated Persuasion

We organise the computational literature on persuasion into three capability classes, ordered by the direction of the information flow relative to the persuasion act:

Capability	Direction	Goal
Descriptive	Observation → Explanation	Explain and measure persuasion in existing content
Simulative	Content → Predicted Response	Predict human persuasive response before it occurs
Generative	Target → Optimised Content	Create maximally persuasive content for a given audience

5.1 Descriptive Capabilities

Descriptive systems explain the mechanics of persuasion in observed content. The core task is understanding what properties of content correlate with persuasive outcomes.

5.1.1 Content Capabilities

Understanding persuasive content requires parsing three intertwined signals:

Content and Logic The quality of arguments, the presence of logical fallacies, the structure of reasoning chains. Computational argumentation mining extracts and classifies these automatically.

Advertisements Visual and multimodal persuasion; predicting ad memorability, click-through, and brand recall from image and video features.

Emotion and Affect Sentiment analysis, emotion recognition, and the mapping of affective content to persuasive outcomes across different demographics.

5.2 Simulative Capabilities

Simulative systems predict how a given person or population will respond to a given message before the message is deployed. This is hard because persuasive effects are moderated by audience characteristics that are rarely directly observed.

Representative lines of work:

- Can Language Models Recognise Convincing Arguments? — testing LLMs as judges of argument quality
- Measuring and Increasing Persuasiveness of Large Language Models [18]
- Memorability prediction from content and context

5.2.1 Single Person

Micro-targeting and personalisation — predicting individual response. Key challenge: the privacy vs. effectiveness trade-off.

5.2.2 Interpersonal Interaction

> Caution
>
> Section under active development.

5.2.3 Societal

- Propaganda detection
- Predicting opinion shifts in populations

5.2.4 Opinion Dynamics

Agent-based models of belief propagation; echo chambers and filter bubbles.

5.2.5 Content Recommendation

Given an audience, select the content that maximises a persuasive objective.

5.2.6 Audience Selection

Given a piece of content, identify the audience for which it will be most persuasive.

5.3 Generative Capabilities

Generative systems produce persuasive content optimised for a particular configuration of **audience**, **time**, **channel**, **sender**, and **topic**. This is the most powerful — and ethically fraught — capability class.

Large language models have dramatically lowered the cost of generating fluent, targeted persuasive text at scale. This raises questions that the research community is only beginning to confront: How persuasive are LLM-generated messages compared to human-written ones? Under what conditions do they outperform human writers? What disclosure obligations do deployers have?

> ⚠ Open Ethical Question
>
> If an AI system can generate a message that is more persuasive than any human writer could produce, targeted at a single individual, delivered via the channel and at the time of greatest receptivity — what governance structures should constrain its deployment?

6 Resources for Persuasion Research

> **Caution**
>
> This chapter catalogues datasets, benchmarks, and toolkits used in computational persuasion research. It is under active development. Suggest a resource via GitHub Issues.

7 The Last Twenty Years: Data, Models, and Behavioural Science at Scale

ℹ Chapter Overview

This chapter traces the most consequential structural change in the study of persuasion since Aristotle: the emergence of digital data and machine learning as instruments for observing, analysing, and ultimately generating persuasive communication at population scale.

7.1 A Before and After

For most of human history, the study of persuasion was limited by what could be directly observed: a speech in the agora, a pamphlet, an advertisement, a laboratory experiment with dozens of undergraduate participants. The mechanisms of attitude change were inferred from small samples, self-reports, and controlled but artificial stimuli. The *reach* of any persuasive act was difficult to measure, and the diversity of human responses to the same message was largely invisible.

The last twenty years have fundamentally altered this situation. Three developments, each building on the last, have together created a new scientific infrastructure for understanding and producing persuasion at scale:

1. **The internet as a data-collection instrument** — the digitisation of everyday behaviour into machine-readable traces.
2. **Machine learning at scale** — the development of methods capable of finding structure in datasets orders of magnitude larger than anything previously studied.
3. **Large Language Models** — systems trained on the accumulated textual output of humanity, capable of generating fluent, contextually appropriate, and persuasive communication.

7.2 The Internet and the Instrumentation of Behaviour

The internet did not merely create a new communication channel — it created a new kind of scientific instrument. Every search query, every click, every purchase, every moment of attention leaves a digital trace: a record of human behaviour generated as a byproduct of ordinary life rather than constructed for scientific purposes [22]. By the early 2010s, the aggregate volume of these traces — what came to be called *big data* — exceeded anything the social sciences had previously encountered.

[22] describe this as enabling observation of "populations and phenomena that were previously difficult or impossible to study." Location data from mobile phones can reveal urban mobility patterns. Search engine queries serve as real-time proxies for public health trends, most famously (and controversially) in the case of Google Flu Trends [23]. Social media posts have been used to infer psychological traits, predict economic indicators, and track the spread of misinformation. In each case, the underlying scientific move is the same: substitute scarce, expensive, purpose-built data collection (surveys, experiments, ethnography) with abundant, cheap, incidentally generated digital traces.

The implications for persuasion research are profound. Rather than measuring attitude change in a laboratory sample of 80 participants, researchers can observe the response of millions of people to the same message in the wild. Rather than constructing artificial stimuli, researchers can study the persuasive effects of actual political advertisements, actual product descriptions, actual public health communications — on their actual audiences.

7.2.1 Digital Traces and the Limits of Measurement

[22] are also careful to note the dangers. Digital data sources were not designed for scientific measurement and bring their own distortions: platform algorithms determine which content is seen, changing affordances alter what traces are generated, and the populations that produce digital data are not representative of humanity [27]. Most fundamentally, human behaviour is not stable under observation: the act of measuring changes what is measured, as platforms that optimise for engagement discover when they shape the very behaviour they are trying to predict.

The engagement-optimisation literature makes this tension vivid. [15] show that maximising user engagement — the primary metric of most digital platforms — does not necessarily maximise user welfare. Platforms can increase the time users spend on a service while simultaneously making them less happy, because the behaviours that are most measurable (clicks, time-on-site, shares) are the ones most subject to impulsive rather than reflective choice. The persuasion effects of platform design are therefore not merely incidental: they are built into the measurement infrastructure itself.

7.3 The Wikipedia Moment: Big Data Meets Social Science

The availability of large digital corpora created an early opportunity to test whether the social sciences could make use of them. Wikipedia, with its comprehensive edit histories and contributor metadata, became a proving ground. [24] showed that large-scale analysis of Wikipedia contributions could reveal geographic imbalances in knowledge production, patterns of conflict and consensus among editors, and the dynamics of public goods contributions at a scale unattainable by any previous method.

Yet [24] also found that this research remained largely **disciplinarily siloed**: studies of Wikipedia tended to answer questions motivated by the structure of the available data rather than by established theoretical questions, and findings were rarely integrated across fields. Big data, it turned out, was not a substitute for theory — it was a complement that required better theories to exploit well.

7.4 The ImageNet Inflection: Machine Learning at Scale

The second development was the maturation of machine learning into a tool capable of finding structure in datasets of unprecedented size. The pivotal moment was the 2012 ImageNet Large Scale Visual Recognition Challenge, in which Krizhevsky, Sutskever, and Hinton demonstrated that a deep convolutional neural network (AlexNet) could reduce image classification error from 26% to 15% — a margin so large that it immediately shifted the direction of the entire field [6, 17].

ImageNet itself — a manually labelled database of over 14 million images across more than 20,000 categories, assembled by Fei-Fei Li and colleagues — made this possible by providing a training set large enough to prevent overfitting in high-capacity models. The lesson generalised immediately: **the combination of large, well-labelled datasets and high-capacity models trained with sufficient compute produced qualitative improvements** in performance on tasks previously considered intractable.

The implications for behavioural research were not immediately obvious, but they became so within a decade. If models trained on millions of images could learn to recognise objects, scenes, and faces, could models trained on millions of human interactions learn to recognise — and produce — persuasion?

7.5 Large Language Models: Persuasion Industrialised

The third development was the emergence of Large Language Models (LLMs): neural networks trained on massive corpora of human-generated text to predict the next token in a sequence. The training corpora are themselves extraordinary artefacts — aggregations of the textual output of human civilisation at a scale never before assembled for scientific or engineering purposes.

- **BERT** [7], introduced by Google in 2018, was pre-trained on the English Wikipedia and the BooksCorpus — roughly 3.3 billion words.
- **GPT-3** [5], introduced by OpenAI in 2020, was trained on Common Crawl, WebText, BooksCorpus, and Wikipedia — roughly 570 GB of filtered text, encompassing hundreds of billions of words of internet content.
- **T5** [31] introduced the encoder-decoder framing that unified text classification, translation, summarisation, and question-answering into a single model architecture trained on the Colossal Clean Crawled Corpus (C4), a 750 GB filtered web corpus.
- Subsequent models — LLaMA [37], Falcon, BLOOM — extended the scale to trillions of tokens and dozens of languages.

The effect of training on text at this scale is qualitative, not merely quantitative. These models acquire not just language statistics but **world knowledge, cultural context, and, critically, an implicit model of how humans communicate, argue, and persuade.** A model trained to predict the next word in a persuasive political speech has, in some sense, learned something about persuasion — not from explicit instruction but from statistical exposure to millions of examples.

LLMs can generate persuasive messages, adapt linguistic style to specific audiences, construct arguments, anticipate counter-arguments, and calibrate emotional tone [3]. Studies have shown that GPT-4-generated persuasive essays are rated as more persuasive than human-written essays by blind evaluators [39], and that AI-generated messages can change opinions on contested political topics [10].

7.6 Large Content and Behaviour Models

The most recent development extends the LLM paradigm to **behavioural data**: not just text, but the full range of human digital traces — clicks, views, purchases, engagement patterns, search queries. Models in this class, which we term Large Content and Behaviour Models (LCBMs) following [19], are trained on joint corpora of content and the behavioural responses that content elicited from real users at scale.

Where LLMs learn *what people say*, LCBMs learn *what people do in response to what they see.* This distinction is significant for persuasion research. A model that has been trained on the engagement patterns of millions of users across millions of pieces of content has, in some sense, learned a generalised model of persuasive effectiveness — what kinds of content, in what contexts, for what audiences, produce what behavioural changes.

[19] demonstrate that LCBMs trained on large-scale content-behaviour corpora significantly outperform content-only models on downstream behavioural prediction tasks, including engagement prediction, sentiment forecasting, and audience response modelling. The intuition is straightforward: knowing that a message was watched, shared, or acted upon by millions of people provides training signal that text alone cannot supply.

7.7 LLMs as Simulators of Human Behaviour

A striking recent trend has been the use of LLMs not as *generators* of persuasion but as *simulators* of human responses to persuasion — in effect, as synthetic subjects for social science experiments.

[2] showed that GPT-3 can be "conditioned" on demographic profiles (age, education, political affiliation, geographic region) to produce responses that mirror the survey responses of those groups with remarkable fidelity — a technique they term "algorithmic fidelity." The implication is that LLMs internalise statistical representations of group-level opinion distributions that can be queried, allowing researchers to generate synthetic survey data for population subgroups that are expensive or difficult to sample directly.

[1] extended this finding, showing that LLMs can replicate the results of classic social science experiments — including Milgram's obedience studies and Kahneman and Tversky's framing effects — without any task-specific training. The models produce not just plausible responses but responses that match the *direction and magnitude* of the original experimental effects.

More directly relevant to persuasion, [4] found that GPT-4 could predict the results of randomised controlled experiments on political persuasion with accuracy comparable to moderately sized human samples. The model was given the treatment descriptions and outcome measures from a battery of persuasion experiments and asked to predict effect sizes — and it succeeded, suggesting that something in its training had captured the regularities of human susceptibility to persuasive messages.

[28] demonstrated that populations of LLM agents, given minimal backstory descriptions, can simulate emergent social dynamics — opinion polarisation, information cascading, social influence — that closely mirror patterns observed in real human populations.

[29] showed that LLMs can capture human personality traits as measured by standardised psychometric instruments (Big Five, MBTI), reproducing known correlations between personality and persuasibility. Models conditioned on high-neuroticism profiles, for example, showed heightened susceptibility to fear-based appeals, consistent with decades of empirical literature on personality and persuasion.

The Limits of LLM Simulation

These findings should be interpreted carefully. LLMs are trained predominantly on text produced by Western, English-speaking, educated, and digitally connected populations — what [**henrich2010weirdest?**] terms the WEIRD bias. Simulation accuracy degrades for populations underrepresented in training data: people over 65, the highly religious, non-English speakers, and communities without significant digital footprints [22]. The claim that LLMs are "human simulators" is therefore a claim that they are simulators of *particular* humans — a limitation with significant implications for the use of LLM simulation in persuasion research.

7.8 What This Means for the Study of Persuasion

The developments traced in this chapter — digital instrumentation, large-scale machine learning, and LLMs — have not merely given persuasion researchers new tools. They have changed the object of study itself.

Persuasion is no longer primarily a dyadic phenomenon (one sender, one receiver, one message). It now operates at the intersection of algorithmic content selection, platform incentive structures, user behavioural data, and industrially-generated content. The mechanisms by which a message reaches a receiver — and the mechanisms by which the receiver's response feeds back to determine what message they see next — are now computational and largely invisible to both sender and receiver.

Understanding persuasion in this environment requires the integration of the classical persuasion literature (elaboration likelihood, attitude change, source credibility) with the emerging literature on algorithmic mediation, behavioural data at scale, and LLM capabilities. That integration is the animating ambition of this review.

Table 7.1: Evolution of the empirical infrastructure for persuasion research.

Era	Primary data source	Primary method	Scale of observation
Pre-digital (–2000)	Surveys, experiments	Statistical inference	Hundreds to thousands
Early digital (2000–2012)	Web traces, social media	Descriptive analytics, NLP	Millions
Big data / deep learning (2012–2020)	Platform logs, images, video	Deep neural networks	Billions
LLM era (2020–)	Multimodal corpora + behavioural signals	Foundation models, LCBMs	Trillions of tokens

8 Future Trends and Unsolved Questions

- **Micro-targeting at LLM scale** — as generation costs approach zero, personalised persuasion will become the norm. What individual rights are at stake?
- **Multi-modal persuasion** — integrating text, image, audio, and video into unified persuasion models.
- **Longitudinal effects** — most studies measure immediate attitude change; long-run effects of AI-generated persuasion are almost completely unstudied.
- **Cross-cultural persuasion** — what works in one cultural context often fails in another; LLMs trained predominantly on English data may encode WEIRD (Western, Educated, Industrialised, Rich, Democratic) persuasion norms.
- **Detection & provenance** — as generative AI scales, detecting AI-generated persuasive content becomes a critical counterbalancing capability.
- **Regulation** — the EU AI Act and proposed US legislation are beginning to address AI-mediated persuasion; the regulatory landscape is rapidly evolving.

i This Review is Living

Entries are added and revised as the field evolves. The source and compiled PDF are always available on GitHub. The PDF updates automatically on every commit via GitHub Actions.

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